



THE SAFETY NET HANDLING MISTAKES IN PYTHON



THE PUNCTURED RICKSHAW STORY

Imagine you're going to school in a rickshaw, and suddenly... PUNCTURE! The tire goes flat! Uh-oh! What happens now? What do we do?

Unexpected problems can happen anytime — even to our programs!



WHAT IS AN EXCEPTION?

An Exception is an unexpected problem that stops our program from working properly.

Just like a rickshaw puncture stops your ride to school, an exception stops your code!



REMEMBER THIS!

- Exception = Surprise problem
- It can make the program stop suddenly
- But don't worry, we can handle it!

THE 'TRY' BLOCK: THE TEST AREA

WHAT IS 'TRY'?

The 'try' block is like testing if you have milk before making chai. We put code that might cause a problem inside 'try' to test it safely!

WE TRY SOMETHING FIRST!

Simple code example:

```
try:  
make_Tea()
```



THE 'EXCEPT' BLOCK: THE BACKUP PLAN

WHAT IS 'EXCEPT'?

'except' is our backup plan! If milk is finished at home, we go to the kirana store to buy more. Easy!

IF FIRST PLAN DOESN'T WORK...

Simple code example:

```
except:  
    go_to_kirana_store()
```



CALCULATOR THAT DOESN'T CRASH!

ZeroDivisionError is like trying to share zero sweets among friends — it just doesn't work!

Here's how we handle it:

try:

```
result = number / divisor
```

```
except ZeroDivisionError:
```

```
print("Oopsie! You can't divide by zero.")
```

With try and except, our calculator stays friendly instead of crashing!



CRASHING

😞 Sad face!

✗ Program stops suddenly

⚠ No warning given

💔 User gets confused

"Oh no! Everything broke!"



HANDLING ERRORS

😊 Smiley face!

⚡ Program keeps running

💛 Friendly warning shown

🎯 "Oopsie, try again!"

NAMEERROR: CALLING A FRIEND WHO ISN'T THERE

A NameError happens when you call a friend who isn't at home! Imagine shouting "Ananya!" but Ananya isn't there. Python gets confused too!

Code example:

```
print(Ananya) # Ananya is not defined
```

Python says: "Oops, I don't know that name!"

Remember: Always introduce your variables before using them!



REAL-WORLD GRACEFUL FAILURE



ATM Example: If an ATM runs out of cash, it shows a friendly message instead of switching off suddenly!

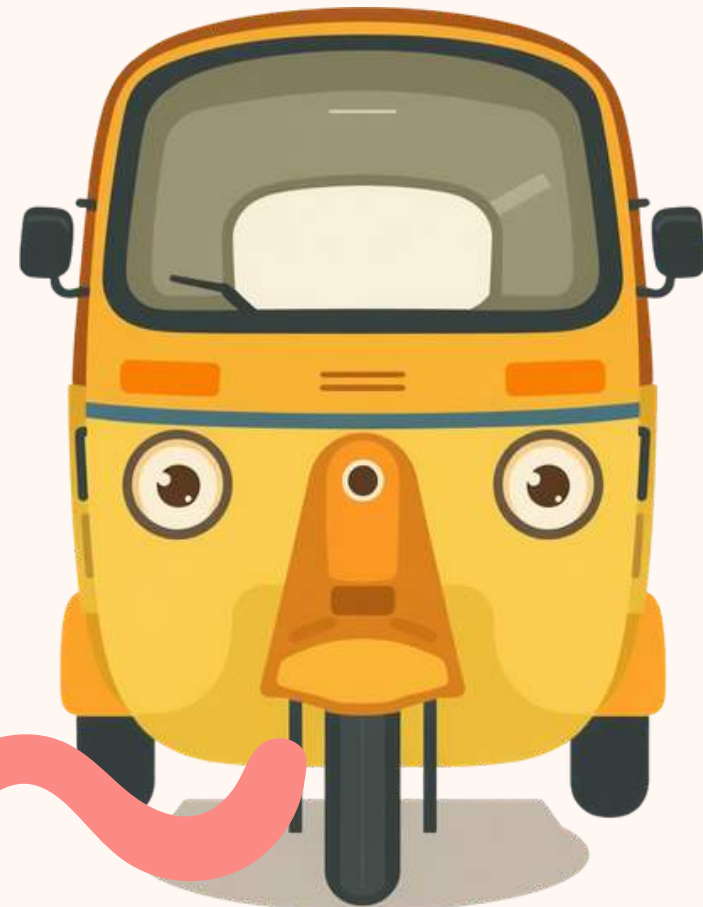
Power Cut Example: During power cuts, the inverter switches on to save the day!

Graceful failure means saying "Oopsie!" nicely instead of stopping suddenly.



QUICK SAFETY NET REMINDER

Remember these key terms to catch errors like a pro! Each one helps keep your Python programs running smoothly.





QUIZ TIME! WHAT'S THE SAFETY NET?

WHAT DOES 'TRY' MEAN?

- A) Test Area ✓
- B) Backup Plan

WHICH IS THE BACKUP PLAN?

- A) try
- B) except ✓

DIVIDING BY ZERO GIVES...

- A) Program crashes
- B) Friendly "Oopsie!" message ✓





SUMMARY: OUR SAFETY NET IN PYTHON



EXCEPTIONS – SURPRISE PROBLEMS

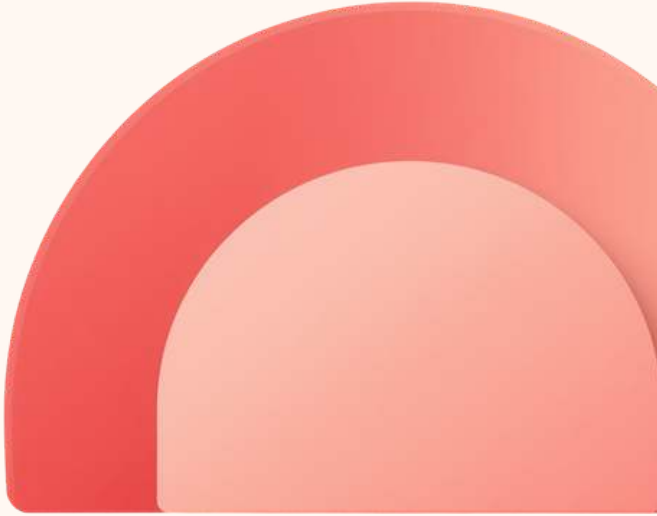
Just like a punctured rickshaw
tire stops your ride, exceptions
are unexpected problems that
can stop our program!

TRY – TEST AREA

Like checking if you have milk
before making Tea, 'try' tests
the code that might cause a
problem.

EXCEPT – BACKUP PLAN

If something goes wrong,
'except' catches the problem
and keeps our program
friendly and safe!



THANK YOU!

Remember, errors are chances to learn! Any questions?





**KEEP PRACTICING
YOUR SAFETY
NETS!**